

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0500

Galion Generating Station 230 kV
February 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0500. The Interconnection Customer proposes a 323.2 MW wind facility interconnecting at the Galion Generating Station 230 kV substation in the Ohio area. The request was received on July 13, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 232 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$11,461,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Portage Lakes / Weedpatch 230 KV	Surrowiec Substation / Weedpatch 500 KV line	119.4	638	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Weedpatch / Portage Lakes 230 KV	Rebuild line with higher-capacity conductor	\$25,408,000	45% (\$11,461,000)
	Total allocated to project		\$11,461,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.